

Expressions and operator order

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Values variables and types

Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Predict then test expressions for Expressions and operator order. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Expressions and operator order. Proposed explanation: “All operations are read left to right regardless of precedence”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Group an expression deliberately

Predict each line. Explain which calculation groups addition before division. What would $(17+5)\%5$ return, and how do `/`, `//` and `%` differ for the supplied positive integers?

```
print((17 + 5) // 5)
print(17 + 5 // 5)
print(17 / 5, 17 // 5, 17 % 5)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 State the intended grouping

Write an expression for three packs of eight counters plus two loose counters. Explain why adding two within the parentheses for every pack describes a different situation.

My reasoning and evidence

6. W2 Predict then test expressions

Predict $16 - 5 - 3$, $16 - (5 - 3)$, and $18 / 3 * 2$. Write the grouping, run each line and record actual results separately.

My reasoning and evidence

7. W3 Explain whole groups

For 17 counters in groups of 5, explain $/$, $//$ and $\%$. Show a multiplication-and-addition check for the complete groups and remainder.

My reasoning and evidence

8. W4 Translate a new situation independently

Four trays hold six tiles each and five tiles are loose. Write and evaluate the total expression. Then pack the total into complete groups of four and find the remainder. Explain why $4 * (6 + 5)$ is wrong for this story.

My reasoning and evidence

9. W5 Check a limit

Explain why a group size of zero must be excluded. State the assumptions that let complete groups and a remainder describe the packing task.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled expressions and operator order evidence sheet to Python readiness portfolio. Show how you can evaluate numeric expressions using python's grouping rules. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Expressions and operator order

Parentheses change an expression. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.

Python 3: `print(3+2*5)`; `print((3+2)*5)`. Inputs are dimensionless for this arithmetic exercise.

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Expressions and operator order

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Use precedence

In $3+2*5$, multiplication happens first.



2. Compute first result

$2*5$ is 10; adding 3 gives 13.



3. Compute grouped result

Parentheses give $3+2=5$ first; multiplying by 5 gives 25.

CHANGE ONE THING

Replace the final multiplier 5 with 4 in both expressions.

Explain the changed result using the matching lesson.

Trace the operation order in both expressions. Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

13. Project portfolio milestone

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Parentheses change an expression. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Parentheses give $3+2=5$ first; multiplying by 5 gives 25.

1. Compute grouped result
2. Use precedence
3. Compute first result

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compute grouped result Parentheses give $3+2=5$ first; multiplying by 5 gives 25. Parentheses explicitly group an operation. Without them, Python

multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Practice 2: explain the deciding evidence

Changed-case practice: Replace the final multiplier 5 with 4 in both expressions. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. The first result is 11; the grouped result is 20.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The first result is 11; the grouped result is 20. $3+2*4=3+8$; $(3+2)*4=5*4$.

Repair a missing building block

Review Values variables and types (python-bridge/2). Goal: Distinguish integer, floating-point, text and Boolean values. Short practice: Predict both print outputs and explain why they differ. Does `count+tag` work, and what happens to each valid output after `count` becomes ten?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
